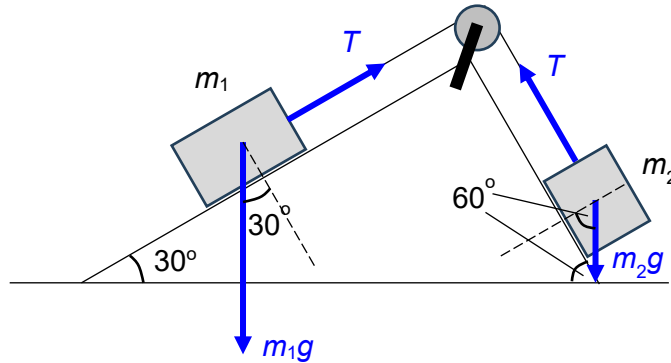


- 7 Two blocks of masses $m_1 = 4.0 \text{ kg}$ and $m_2 = 1.0 \text{ kg}$ are connected by a cord of negligible mass that passes over a frictionless pulley of negligible mass. The blocks slide on frictionless planes inclined at angles $\theta_1 = 30^\circ$ and $\theta_2 = 60^\circ$.



What is the tension in the cord?

A 2.3 N

B 5.8 N

C 8.0 N

D 10.7 N

Ans: D

Apply $F = ma$ to both masses:

$$m_1g \sin 30^\circ - T = m_1a \quad \text{---(1)}$$

$$T - m_2g \sin 60^\circ = m_2a \quad \text{---(2)}$$

Sub. T from (2) into (1):

$$m_1g \sin 30^\circ - (m_2g \sin 60^\circ + m_2a) = m_1a$$

$$(m_1 + m_2)a = g(m_1 \sin 30^\circ - m_2 \sin 60^\circ)$$

$$(4.0 + 1.0)a = 9.81(4.0 \sin 30^\circ - 1.0 \sin 60^\circ)$$

$$a = 2.22 \text{ m s}^{-2}$$

Sub. Into (2):

$$T - 9.81 \sin 60^\circ = 2.22$$

$$T = 10.7 \text{ N}$$